

Win-1 * R1 * R2 * R3 * R4 * R5 * Win-3 * R6 * Win-2 *

R1(config)#ip route 223.0.50.0 255.255.255.0 12.0.0.2

R1(config)#exit

R1#sh

*Aug 25 02:12:32.165: %SYS-5-CONFIG_I: Configured from console by console

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 12.0.0.0/8 is directly connected, Ethernet0/1
L 12.0.0.1/32 is directly connected, Ethernet0/1
S 23.0.0.0/8 [1/0] via 12.0.0.2
S 24.0.0.0/8 [1/0] via 12.0.0.2
S 26.0.0.0/8 [1/0] via 12.0.0.2
S 32.0.0.0/8 [1/0] via 12.0.0.2
S 45.0.0.0/8 [1/0] via 12.0.0.2
S 172.36.0.0/16 [1/0] via 12.0.0.2
192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.50.20.0/24 is directly connected, Ethernet0/0
L 192.50.20.1/32 is directly connected, Ethernet0/0
S 223.0.50.0/24 [1/0] via 12.0.0.2

R1#

R1#

R1#

*Aug 25 02:13:03.167: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)

R1#

```
Terminal
Win-1 * R1 * R2 * R3 * R4 * R5 * Win-3 * R6 * Win-2 *
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
ia - IS-IS inter area, * - candidate default, U - per-user sta
tic route
o - ODR, P - periodic downloaded static route, H - NHRP, l - L
ISP
a - application route
+ - replicated route, % - next hop override, p - overrides fro
m PFR
Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
12.0.0.0/8 is directly connected, Ethernet0/0
12.0.0.2/32 is directly connected, Ethernet0/0
23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
23.0.0.0/8 is directly connected, Ethernet0/2
23.0.0.1/32 is directly connected, Ethernet0/2
24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
24.0.0.0/8 is directly connected, Ethernet0/3
24.0.0.1/32 is directly connected, Ethernet0/3
26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
26.0.0.0/8 is directly connected, Ethernet0/1
26.0.0.1/32 is directly connected, Ethernet0/1
32.0.0.0/8 [1/0] via 23.0.0.2
45.0.0.0/8 [1/0] via 24.0.0.2
172.36.0.0/16 [1/0] via 23.0.0.2
192.50.20.0/24 [1/0] via 12.0.0.1
223.0.50.0/24 [1/0] via 26.0.0.2
R2#
R2#
R2#
```

```
Terminal
Win-1 x R1 x R2 x R3 x R4 x R5 x Win-3 x R6 x Win-2 x

down
Ethernet0/3 unassigned YES TFTP down
down
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
ia - IS-IS inter area, * - candidate default, U - per-user sta
tic route
o - ODR, P - periodic downloaded static route, H - NHRP, l - L
ISP
a - application route
+ - replicated route, % - next hop override, p - overrides fro
m Pfr

Gateway of last resort is not set

S 12.0.0.0/8 [1/0] via 23.0.0.1
23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 23.0.0.0/8 is directly connected, Ethernet0/0
L 23.0.0.2/32 is directly connected, Ethernet0/0
S 24.0.0.0/8 [1/0] via 23.0.0.1
S 26.0.0.0/8 [1/0] via 23.0.0.1
32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 32.0.0.0/8 is directly connected, Ethernet0/1
L 32.0.0.1/32 is directly connected, Ethernet0/1
S 45.0.0.0/8 [1/0] via 23.0.0.1
S 172.36.0.0/16 [1/0] via 32.0.0.2
S 192.50.20.0/24 [1/0] via 23.0.0.1
S 223.0.50.0/24 [1/0] via 23.0.0.1
R3#
```

```
Terminal
Win-1 * R1 * R2 * R3 * R4 * R5 * Win-3 * R6 * Win-2 *
down
Ethernet0/3 unassigned YES TFTP down
down
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
ia - IS-IS inter area, * - candidate default, U - per-user sta
tic route
o - ODR, P - periodic downloaded static route, H - NHRP, l - L
ISP
a - application route
+ - replicated route, % - next hop override, p - overrides fro
m Pfr
Gateway of last resort is not set
S 12.0.0.0/8 [1/0] via 24.0.0.1
S 23.0.0.0/8 [1/0] via 24.0.0.1
24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 24.0.0.0/8 is directly connected, Ethernet0/0
L 24.0.0.2/32 is directly connected, Ethernet0/0
S 26.0.0.0/8 [1/0] via 24.0.0.1
S 32.0.0.0/8 [1/0] via 24.0.0.1
45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 45.0.0.0/8 is directly connected, Ethernet0/1
L 45.0.0.1/32 is directly connected, Ethernet0/1
S 172.36.0.0/16 [1/0] via 45.0.0.2
S 192.50.20.0/24 [1/0] via 24.0.0.1
S 223.0.50.0/24 [1/0] via 24.0.0.1
R4#
```

```
Terminal
Win-1 * R1 * R2 * R3 * R4 * R5 * Win-3 * R6 * Win-2 *
down
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
ia - IS-IS inter area, * - candidate default, U - per-user sta
tic route
o - ODR, P - periodic downloaded static route, H - NHRP, l - L
ISP
a - application route
+ - replicated route, % - next hop override, p - overrides fro
Pfr
>
Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 45.0.0.1
S    23.0.0.0/8 [1/0] via 45.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      32.0.0.0/8 is directly connected, Ethernet0/1
L      32.0.0.2/32 is directly connected, Ethernet0/1
45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      45.0.0.0/8 is directly connected, Ethernet0/2
L      45.0.0.2/32 is directly connected, Ethernet0/2
172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.36.0.0/16 is directly connected, Ethernet0/0
L      172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 45.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
R5#
```

```
Terminal
Win-1 x R1 x R2 x R3 x R4 x R5 x Win-3 x R6 x Win-2 x
Ethernet0/3          unassigned      YES TFTP    down
down
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B -
BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
       ia - IS-IS inter area, * - candidate default, U - per-user sta
tic route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - L
ISP
       a - application route
       + - replicated route, % - next hop override, p - overrides fro
n Pfr

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
     26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        26.0.0.0/8 is directly connected, Ethernet0/1
L        26.0.0.2/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
     223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C        223.0.50.0/24 is directly connected, Ethernet0/0
L        223.0.50.1/32 is directly connected, Ethernet0/0
R6#
R6#
```

```
Terminal
Win-1 * R1 * R2 * R3 * R4 * R5 * Win-3 * R6 * Win-2 *

C:\WINDOWS\system32\cmd.exe
'ip' is not recognized as an internal or external command,
operable program or batch file.
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 223.0.50.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 223.0.50.1

C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=3ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_

Start [Taskbar icons] C:\WINDOWS\system32 3:53 AM
```

Terminal

Win-1 * R1 * R2 * R3 * R4 * R5 * Win-3 * R6 * Win-2 *

C:\WINDOWS\system32\cmd.exe

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address : 172.36.0.10
Subnet Mask : 255.255.0.0
Default Gateway : 172.36.0.1

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=6ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time=3ms TTL=124

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 6ms, Average = 3ms

C:\Documents and Settings\user>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time=3ms TTL=124
Reply from 192.50.20.100: bytes=32 time=3ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124

Ping statistics for 192.50.20.100:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>_

Start C:\WINDOWS\system32\cmd.exe

3:51 AM

```
Terminal
Win-1 * R1 * R2 * R3 * R4 * R5 * Win-3 * R6 * Win-2 *

C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ncpa.cpl

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=2ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=2ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>
```